

1. Dataset

This project uses the *Construction Site Safety dataset* from Roboflow, which contains approximately 700 annotated images of construction environments. The dataset includes bounding box labels for PPE-related classes such as hard hats, safety vests, masks, and non-compliance categories (e.g., NO-Hardhat). The dataset is used for object detection tasks and is split into training, validation, and test sets.

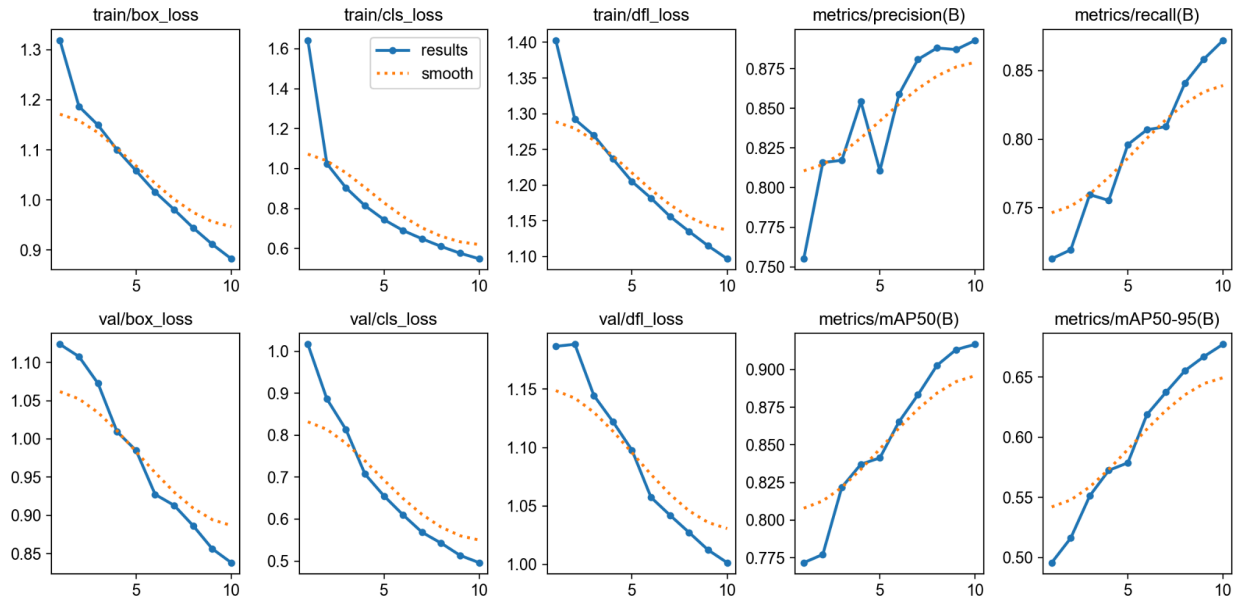
2. Model and Method

A YOLOv8 (Ultralytics) object detection model was used as the baseline CNN for PPE detection. The model was fine-tuned on the dataset using transfer learning with pretrained weights ([yolov8n.pt](#)). Training was performed for 10 epochs with an image size of 640.

3. Results

The model achieved the following performance:

- mAP@0.5: **0.92**
- mAP@0.5–0.95: **0.67**
- Precision: **0.89**
- Recall: **0.87**



4. CNN / NLP / RL Progress

ASafeSight: Week 2 Checkpoint Report

- **CNN**: YOLOv8 baseline successfully trained and evaluated
- **NLP**: Basic module created to generate textual safety alerts (e.g., “Worker without hardhat detected”)
- **RL**: Initial stub implemented to adjust detection thresholds dynamically

Members:

Magat Jr., Gabriel B.

Limpin, Paulo Miguel R.

Pascua, Charles Orville G.